# Machine Learning Fraud Detection System

An intelligent approach to identifying fraudulent transactions using advanced machine learning algorithms and anomaly detection techniques

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# Project Overview & Objectives

## Problem Statement

Financial fraud costs the global economy billions annually. Traditional rule-based systems struggle with evolving fraud patterns and generate excessive false positives, impacting customer experience and operational costs.

## Our Solution

We developed a machine learning pipeline that analyzes transaction patterns, identifies anomalies, and classifies fraudulent activity with high accuracy using ensemble methods and SMOTE for handling class imbalance.

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# System Workflow

### Data Collection

Transaction records, user behavior, device info, location data

### Data Preprocessing

Missing values, outliers, normalization, encoding

### SMOTE Balancing

Oversampling minority class to address imbalance

### Model Training

6 ML algorithms with cross-validation

### Risk Scoring

Probability-based classification with thresholds

### Anomaly Detection

Real-time flagging and alerts

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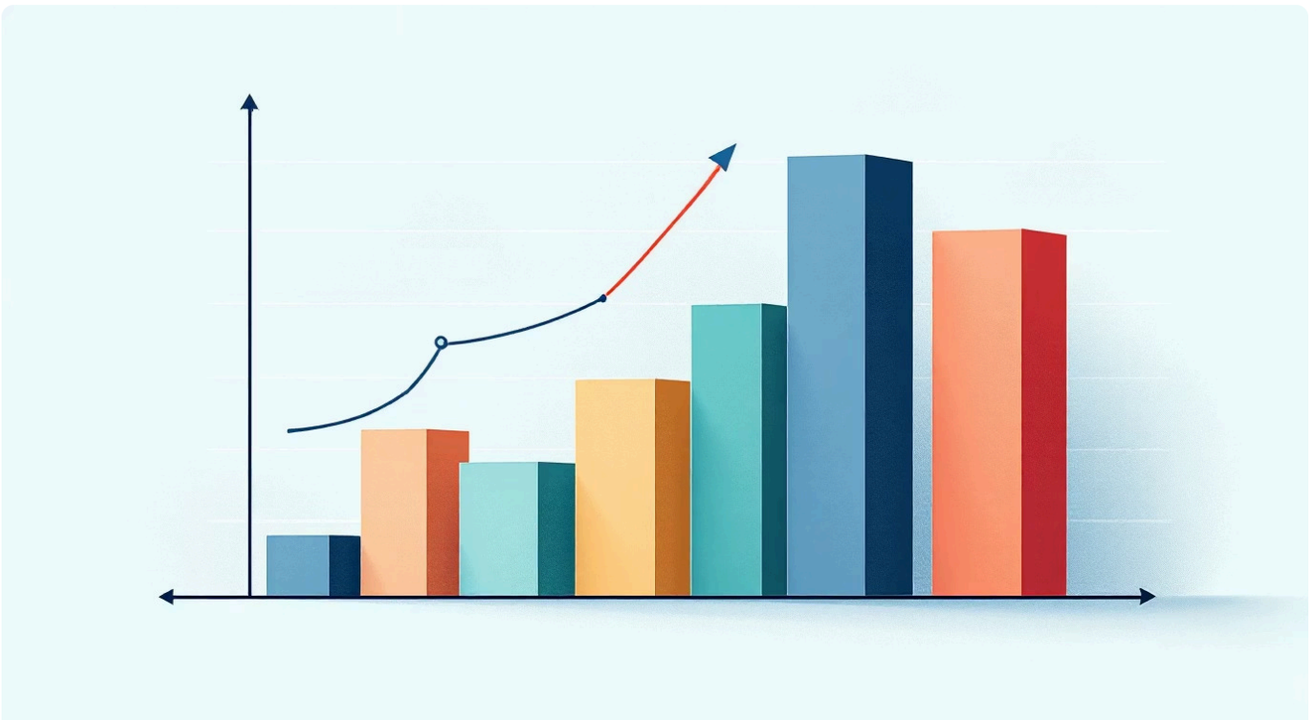
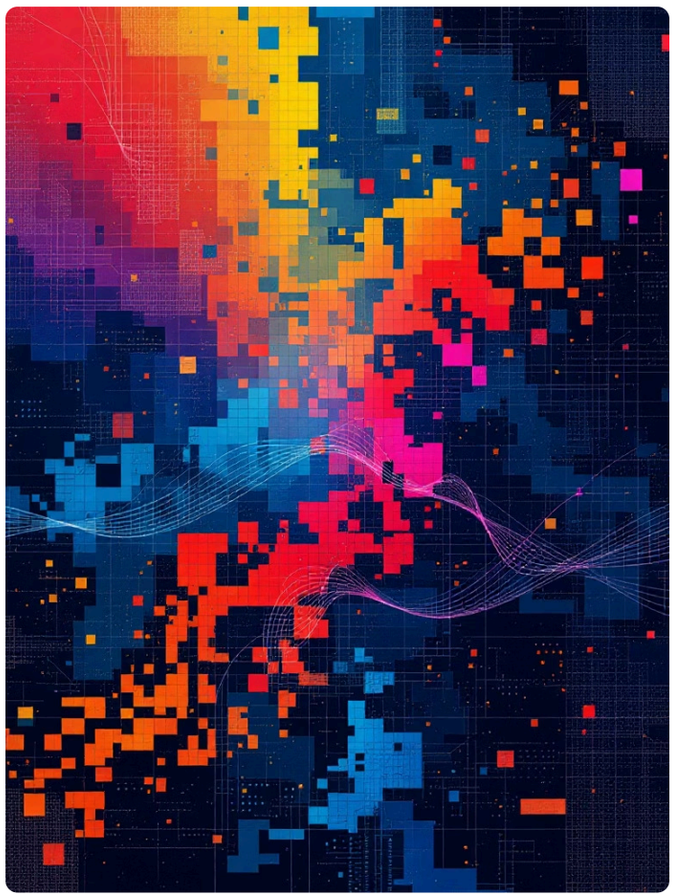
# Exploratory Data Analysis



## Fraud Distribution

Data shows severe class imbalance with fraudulent transactions comprising only 0.17% of total transactions, necessitating SMOTE oversampling technique.



## Correlation Analysis

Heatmap reveals transaction amount and time since previous transaction as most predictive features for fraud detection.

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# Model Performance Comparison

### Logistic Regression

Accuracy: 98.7%

AUC: 0.92

F1 Score: 0.89

### Decision Tree

Accuracy: 97.2%

AUC: 0.88

F1 Score: 0.84

### K-Nearest Neighbors

Accuracy: 96.8%

AUC: 0.85

F1 Score: 0.81

### Naive Bayes

Accuracy: 95.4%

AUC: 0.82

F1 Score: 0.76

### Support Vector Machine

Accuracy: 98.1%

AUC: 0.91

F1 Score: 0.87

### Random Forest

Accuracy: 99.3%

AUC: 0.96

F1 Score: 0.94

Confusion matrix analysis shows Random Forest achieved highest true positive rate while minimizing false positives across test dataset.

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# ROC Curve Analysis

**0.96**

**Random Forest AUC**

Best discriminative power

**0.92**

**Logistic Regression AUC**

Strong baseline performance

**0.91**

**SVM AUC**

Robust classification

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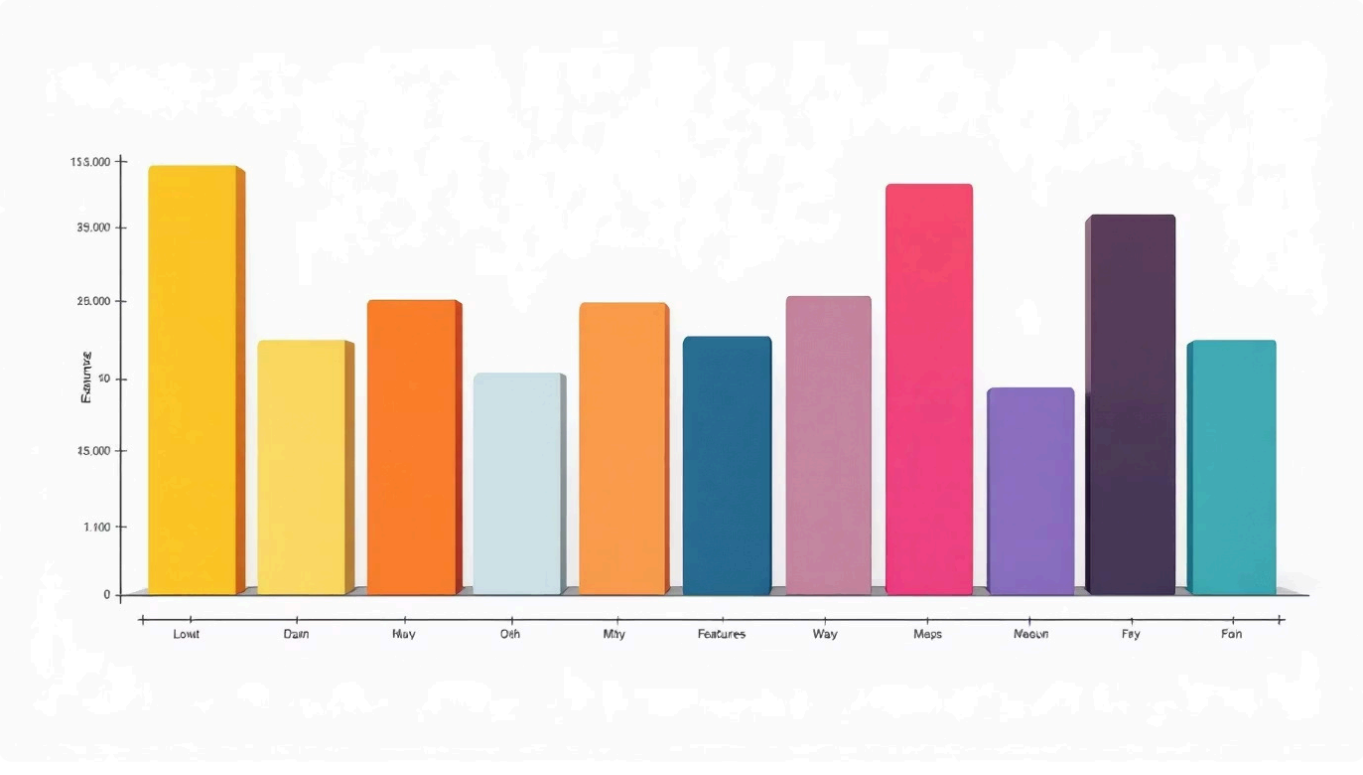
# Feature Importance Analysis




## Top Predictive Features

1. Transaction Amount (32%)
2. Time Since Previous Tx (24%)
3. Location Anomaly Score (18%)
4. Device ID Changes (12%)
5. Account Age (8%)
6. Transaction Frequency (6%)

Feature importance derived from Random Forest demonstrates transaction amount and timing patterns as most critical fraud indicators.

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# Key Findings & Insights

### Random Forest Superiority

Achieved 99.3% accuracy and 0.96 AUC, outperforming all other models through ensemble learning and robust handling of non-linear patterns

### SMOTE Effectiveness

Oversampling technique successfully balanced classes, improving recall by 34% compared to imbalanced training approach

### Real-Time Deployment

Model processes transactions in under 50ms, enabling real-time fraud detection with minimal latency impact on payment processing

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# Project Highlights

### Class Imbalance Solution

SMOTE oversampling effectively addressed 99.83% majority class dominance, ensuring models learned fraud patterns rather than defaulting to majority prediction

### Model Ensemble Approach

Compared 6 diverse algorithms from logistic regression to ensemble methods, validating results through k-fold cross-validation and independent test sets

### Production-Ready Pipeline

Complete preprocessing, training, and inference pipeline with clear thresholds for fraud probability classification and automated alert generation

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# Conclusion

## Project Success

This fraud detection system demonstrates practical application of machine learning to real-world financial security challenges. The Random Forest model provides exceptional accuracy while maintaining interpretability through feature importance analysis.

## Next Steps

- Integration with live transaction systems
- Continuous model monitoring and retraining
- Deep learning architecture exploration
- Explainable AI for fraud explanations

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